Year 13 Maths - Further applications of calculus

Similar questions - Parametric equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has parametric equations $x = t + \frac{3}{t}$, $y = t - \frac{3}{t}$ for $t \neq 0$.

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of t, in its simplest form. **[4]**

(b) Explain why the curve has no stationary points, and find the coordinates of the points where the tangent is parallel to the y-axis. **[4]**

(c) By considering $x+y$ and $x-y$, find a Cartesian equation of the curve. **[3]**

2 A curve has parametric equations $x = \frac{1}{t}$, $y = 2t$ for $t \neq 0$. The point P has coordinates $\left(\frac{1}{p}, 2p\right)$ where $p > 0$.

(a) Show that the tangent at P has equation $y = -2p^2x + 4p$. **[4]**

(b) The tangent at P meets the x-axis at A and the normal at P meets the x-axis at B. Show that $PA : PB = 1 : 2p^2$. **[7]**

3 The curve C has parametric equations $x = 2k + (2k-1)\sin t$, $y = 3 + (k+2)\cos t$ for $0 \leqslant t < 2\pi$, where k is a constant.

(a) Find a Cartesian equation of C. You do not need to simplify it. **[2]**

(b) Given that C is a circle, find the two possible values of k and state the radius and centre of the circle in each case. **[5]**

Question 1

(a) $\frac{dy}{dt} = 1 + \frac{3}{t^2}$, $\frac{dx}{dt} = 1 - \frac{3}{t^2}$, so $\frac{dy}{dx} = \frac{t^2+3}{t^2-3}$

(b) The numerator $t^2+3 > 0$ is never zero, so $\frac{dy}{dx} \neq 0$. The tangent is vertical when $t^2 = 3$: $t = \sqrt{3}$ gives $(2\sqrt{3}, 0)$ and $t = -\sqrt{3}$ gives $(-2\sqrt{3}, 0)$

(c) $x+y = 2t$ and $x-y = \frac{6}{t}$, so $(x+y)(x-y) = 12$, i.e. $x^2 - y^2 = 12$

Question 2

(a) $\frac{dy}{dx} = \frac{2}{-1/t^2} = -2t^2$, at P it is $-2p^2$. $y - 2p = -2p^2\left(x - \frac{1}{p}\right)$ gives $y = -2p^2x + 4p$

(b) A: $y = 0$, $x = \frac{2}{p}$. Normal gradient $\frac{1}{2p^2}$: $y - 2p = \frac{1}{2p^2}\left(x - \frac{1}{p}\right)$; at $y = 0$, $x = \frac{1}{p} - 4p^3$ giving B. $PA^2 = \left(\frac{1}{p}\right)^2 + 4p^2 = \frac{1+4p^4}{p^2}$. $PB^2 = (4p^3)^2 + 4p^2 = 4p^2(1+4p^4)$. So $\frac{PA^2}{PB^2} = \frac{1}{4p^4}$, and $PA : PB = 1 : 2p^2$

Question 3

(a) $\frac{(x-2k)^2}{(2k-1)^2} + \frac{(y-3)^2}{(k+2)^2} = 1$

(b) A circle needs $(2k-1)^2 = (k+2)^2$, so $2k-1 = k+2$ or $2k-1 = -(k+2)$: $k = 3$ or $k = -\frac{1}{3}$. For $k = 3$: radius 5, centre $(6, 3)$. For $k = -\frac{1}{3}$: radius $\frac{5}{3}$, centre $\left(-\frac{2}{3}, 3\right)$